Exercises: Duality Theory and KKT Conditions – Answer Key

Mathematical Foundations for Machine Learning

Lagrangian Formulation and KKT Conditions

Exercise 1.1 [Easy]
Consider the optimization problem

$$\min_{x \in \mathbb{R}^2} x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 \geq 2.$$

(a) Rewrite the constraint in the standard form $g(x) \leq 0$.
(b) Write down the Lagrangian $\mathcal{L}(x, \lambda)$.
(c) State all four KKT conditions for this problem explicitly (do not solve them yet).

Solution

(a) The constraint $x_1 + x_2 \geq 2$ becomes $g(x) = 2 - x_1 - x_2 \leq 0$.
(b) The Lagrangian is:

$$\mathcal{L}(x, \lambda) = x_1^2 + x_2^2 + \lambda(2 - x_1 - x_2)$$

with $\lambda \geq 0$.
(c) The four KKT conditions are:

1. *Stationarity:*

$$\frac{\partial \mathcal{L}}{\partial x_1} = 2x_1 - \lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial x_2} = 2x_2 - \lambda = 0.$$

2. *Primal feasibility:*

$$2 - x_1 - x_2 \leq 0.$$

3. *Dual feasibility:*

$$\lambda \geq 0.$$

4. *Complementary slackness:*

$$\lambda(2 - x_1 - x_2) = 0.$$

Exercise 1.2 [Easy]
Consider the problem from Exercise 1.1. A candidate solution is $x^* = (1, 1)$ with multiplier $\lambda^* = 2$.

(a) Verify that all four KKT conditions are satisfied at (x^*, λ^*).

(b) Is the objective function convex? Is the feasible set convex? What can you conclude about x^*?

(c) Give an interpretation of the value of $\lambda^* = 2$: if the constraint is changed to $x_1 + x_2 \geq 2 + \epsilon$ for small $\epsilon > 0$, by approximately how much does the optimal objective value change?

Solution

(a) Checking each condition:
Stationarity: $2(1) - 2 = 0$ ✓ and $2(1) - 2 = 0$ ✓
Primal feasibility: $2 - 1 - 1 = 0 \leq 0$ ✓
Dual feasibility: $\lambda^* = 2 \geq 0$ ✓
Complementary slackness: $2 \cdot (2 - 1 - 1) = 2 \cdot 0 = 0$ ✓
All four conditions hold.
(b) The objective $f(x) = x_1^2 + x_2^2$ has Hessian $H = 2I \succ 0$, so it is strictly convex. The feasible set $\{x : x_1 + x_2 \geq 2\}$ is a half-space, which is convex. Since the problem is convex and the KKT conditions are satisfied, $x^* = (1, 1)$ is the global minimizer with optimal value $f(x^*) = 2$.
(c) The Lagrange multiplier λ^* measures the sensitivity of the optimal value to perturbations of the constraint. Changing the right-hand side from 2 to $2 + \epsilon$ changes the optimal value by approximately $\lambda^* \cdot \epsilon = 2\epsilon$. So the optimal value increases to approximately $2 + 2\epsilon$. One can verify: the new optimum is $x^* = (1 + \epsilon/2, 1 + \epsilon/2)$ with objective $(1 + \epsilon/2)^2 + (1 + \epsilon/2)^2 = 2 + 2\epsilon + \epsilon^2/2 \approx 2 + 2\epsilon$.

Exercise 1.3 [Medium]
Formulate an optimization problem and use the KKT conditions to find the point in the set

$$M = \{x \in \mathbb{R}^2 : x_1^2 + x_2 \leq 4,\ x_1 - x_2 \leq 0\}$$

closest to the point $p = (0, 5)$, using the squared Euclidean distance as objective.

(a) Write the optimization problem in standard form.

(b) Write the Lagrangian and all KKT conditions.

(c) Use complementary slackness to determine which constraints are active, and solve for the optimal point.

(d) Verify that your solution is indeed a global minimizer.

Solution

(a) The problem is:

$$\min_{x\in\mathbb{R}^2} x_1^2 + (x_2-5)^2 \quad \text{s.t.} \quad g_1(x) = x_1^2 + x_2 - 4 \le 0, \quad g_2(x) = x_1 - x_2 \le 0.$$

(b) Lagrangian:

$$\mathcal{L} = x_1^2 + (x_2-5)^2 + \lambda_1(x_1^2 + x_2 - 4) + \lambda_2(x_1 - x_2)$$

KKT conditions:

- Stationarity: $2x_1 + 2\lambda_1 x_1 + \lambda_2 = 0$ and $2(x_2-5) + \lambda_1 - \lambda_2 = 0$
- Primal feasibility: $x_1^2 + x_2 \le 4$ and $x_1 - x_2 \le 0$
- Dual feasibility: $\lambda_1, \lambda_2 \ge 0$
- Complementary slackness: $\lambda_1(x_1^2 + x_2 - 4) = 0$ and $\lambda_2(x_1 - x_2) = 0$

(c) We consider cases via complementary slackness.
Case 1: Both constraints inactive ($\lambda_1 = \lambda_2 = 0$). Stationarity gives $x_1 = 0, x_2 = 5$. Check feasibility: $0 + 5 = 5 > 4$. Infeasible.
Case 2: Only g_1 active ($\lambda_2 = 0$, $x_1^2 + x_2 = 4$). From stationarity: $2x_1(1+\lambda_1) = 0$. If $x_1 = 0$: then $x_2 = 4$. Second equation: $2(4-5) + \lambda_1 = 0 \Rightarrow \lambda_1 = 2 > 0$ ✓. Check g_2: $0 - 4 = -4 \le 0$ ✓. Candidate: $x^* = (0,4)$ with $\lambda_1 = 2, \lambda_2 = 0$.
Case 3: Only g_2 active ($\lambda_1 = 0$, $x_1 = x_2$). Stationarity: $2x_1 + \lambda_2 = 0$ and $2(x_1 - 5) - \lambda_2 = 0$. Adding: $4x_1 - 10 = 0 \Rightarrow x_1 = 5/2$, $\lambda_2 = -5 < 0$. Dual infeasible.
Case 4: Both active ($x_1^2 + x_2 = 4$, $x_1 = x_2$). Then $x_1^2 + x_1 = 4 \Rightarrow x_1 = \frac{-1+\sqrt{17}}{2} \approx 1.56$. Stationarity gives $\lambda_2 = -2x_1(1+\lambda_1)$ and $2(x_2-5) + \lambda_1 - \lambda_2 = 0$. Working through: one finds λ_1 or λ_2 becomes negative. Dual infeasible.
The unique KKT point is $x^* = (0,4)$.
(d) The objective is convex (sum of convex functions). The constraint g_1 is convex (Hessian of $x_1^2 + x_2$ is $\mathrm{diag}(2,0) \succeq 0$), and g_2 is affine. So the problem is convex. Since KKT conditions are satisfied, $x^* = (0,4)$ is a global minimizer with optimal value $0 + 1 = 1$.

Exercise 1.4 [Medium]
Consider the problem of minimizing the squared norm on the probability simplex:

$$\min_{x\in\mathbb{R}^n} \sum_{i=1}^n x_i^2 \quad \text{s.t.} \quad \sum_{i=1}^n x_i = 1, \quad x_i \ge 0\ \forall i.$$

(a) Write the Lagrangian with multiplier μ for the equality constraint and $\lambda_i \ge 0$ for the inequality constraints $-x_i \le 0$.

(b) Write the KKT conditions and use complementary slackness to argue that $x_i^* = \frac{1}{n}$ for all i.

(c) What is the optimal value? How does it behave as $n \to \infty$? Give an interpretation in the context of diversification in portfolio optimization.

Solution

(a) The Lagrangian is:

$$\mathcal{L}(x, \lambda, \mu) = \sum_{i=1}^{n} x_i^2 + \mu \left(\sum_{i=1}^{n} x_i - 1 \right) + \sum_{i=1}^{n} \lambda_i(-x_i)$$

with $\lambda_i \geq 0$ for all i.
(b) *Stationarity:* $\frac{\partial \mathcal{L}}{\partial x_i} = 2x_i + \mu - \lambda_i = 0$ for all i.
Complementary slackness: $\lambda_i x_i = 0$ for all i.
Suppose $x_i > 0$ for some i. Then $\lambda_i = 0$ by complementary slackness, so $x_i = -\mu/2$. This value is the same for all indices where $x_i > 0$. If any $x_j = 0$, then stationarity gives $\lambda_j = \mu$. Since $\lambda_j \geq 0$ and $x_i = -\mu/2 > 0$ requires $\mu < 0$, we get $\lambda_j = \mu < 0$, contradicting dual feasibility.
Therefore all $x_i > 0$, all $\lambda_i = 0$, and $x_i = -\mu/2$ for all i. From $\sum x_i = 1$: $n \cdot (-\mu/2) = 1 \Rightarrow \mu = -2/n$. Thus $x_i^* = 1/n$ for all i.
(c) The optimal value is $\sum_{i=1}^{n} (1/n)^2 = 1/n$. As $n \to \infty$, the optimal value $\to 0$. In portfolio optimization, $\|x\|^2$ is related to portfolio variance (when assets are uncorrelated with equal variance). The result $x_i = 1/n$ is the *equal-weight portfolio*, and $1/n \to 0$ shows that diversification across more assets reduces risk.

Exercise 1.5 [Hard]
Ridge regression as constrained optimization. Consider the problem

$$\min_{\theta \in \mathbb{R}^D} \|Y - X\theta\|_2^2 \quad \text{s.t.} \quad \|\theta\|_2^2 \leq t,$$

where $X \in \mathbb{R}^{N \times D}$, $Y \in \mathbb{R}^N$, and $t > 0$.

(a) Write the Lagrangian and derive the KKT conditions.

(b) From the stationarity condition, show that the optimal θ^* satisfies

$$\theta^* = (X^T X + \lambda I)^{-1} X^T Y$$

for some $\lambda \geq 0$, and explain why $X^T X + \lambda I$ is always invertible for $\lambda > 0$.

(c) Use complementary slackness to describe when $\lambda = 0$ (no regularization) vs. $\lambda > 0$ (active regularization). Relate this to the OLS estimator $\hat{\theta}_{\text{OLS}} = (X^T X)^{-1} X^T Y$.

(d) Prove that this constrained problem is convex and that strong duality holds (verify Slater's condition).

Solution

(a) The Lagrangian is:

$$\mathcal{L}(\theta, \lambda) = \|Y - X\theta\|^2 + \lambda(\|\theta\|^2 - t), \quad \lambda \geq 0.$$

KKT conditions:

- Stationarity: $-2X^T(Y - X\theta) + 2\lambda\theta = 0$
- Primal feasibility: $\|\theta\|^2 \leq t$
- Dual feasibility: $\lambda \geq 0$
- Complementary slackness: $\lambda(\|\theta\|^2 - t) = 0$

(b) The stationarity condition gives:

$$X^T X\theta + \lambda\theta = X^T Y \quad \Longrightarrow \quad (X^T X + \lambda I)\theta = X^T Y.$$

For $\lambda > 0$: the matrix $X^T X + \lambda I$ is positive definite because $X^T X \succeq 0$ and $\lambda I \succ 0$, so their sum is positive definite and hence invertible. Thus:

$$\theta^* = (X^T X + \lambda I)^{-1} X^T Y.$$

(c) By complementary slackness, $\lambda(\|\theta^*\|^2 - t) = 0$.
Case $\lambda = 0$*:* The constraint is inactive. Then $\theta^* = (X^T X)^{-1} X^T Y = \hat{\theta}_{\text{OLS}}$ (assuming $X^T X$ is invertible). This happens when the unconstrained OLS solution already satisfies $\|\hat{\theta}_{\text{OLS}}\|^2 \leq t$, i.e., the constraint ball is large enough to contain the OLS solution.
Case $\lambda > 0$*:* The constraint is active: $\|\theta^*\|^2 = t$. The solution lies on the boundary of the ball, and λ shrinks the coefficients toward zero. Larger λ corresponds to smaller t (tighter constraint), producing more regularization.
(d) *Convexity:* The objective $\|Y - X\theta\|^2$ is convex in θ (it is a non-negative definite quadratic since its Hessian is $2X^T X \succeq 0$). The constraint set $\{\theta : \|\theta\|^2 \leq t\}$ is convex (it is a Euclidean ball). So the problem is convex.
Slater's condition: We need a strictly feasible point, i.e., θ with $\|\theta\|^2 < t$. Take $\theta = 0$: $\|0\|^2 = 0 < t$. So Slater's condition holds and strong duality follows.

Dual Function and Dual Problem

Exercise 2.1 **[Easy]**
Consider the one-dimensional problem

$$\min_{x \in \mathbb{R}} x^2 \quad \text{s.t.} \quad x \geq 2.$$

(a) Rewrite in standard form and compute the Lagrangian.
(b) Derive the dual function $q(\lambda)$ by minimizing the Lagrangian over x.
(c) State the dual problem and solve it. What is d^*? What is p^*? Is there a duality gap?

Solution

(a) Standard form: $\min x^2$ s.t. $g(x) = 2 - x \leq 0$. Lagrangian:

$$\mathcal{L}(x, \lambda) = x^2 + \lambda(2 - x), \quad \lambda \geq 0.$$

(b) Minimize over x:

$$\frac{\partial \mathcal{L}}{\partial x} = 2x - \lambda = 0 \quad \Longrightarrow \quad x = \frac{\lambda}{2}.$$

Substitute back:

$$q(\lambda) = \left(\frac{\lambda}{2}\right)^2 + \lambda\left(2 - \frac{\lambda}{2}\right) = \frac{\lambda^2}{4} + 2\lambda - \frac{\lambda^2}{2} = 2\lambda - \frac{\lambda^2}{4}.$$

(c) Dual problem: $\max_{\lambda \geq 0} \; 2\lambda - \frac{\lambda^2}{4}$.
This is a concave quadratic. Setting $q'(\lambda) = 2 - \lambda/2 = 0$ gives $\lambda^* = 4$. Thus $d^* = q(4) = 8 - 4 = 4$.
The primal optimum is $x^* = 2$ with $p^* = 4$.
Since $d^* = p^* = 4$, there is **no duality gap**. Strong duality holds (as expected: convex problem, Slater's condition satisfied with e.g. $x = 3$).

Exercise 2.2 [Medium]
Dual of least-squares with linear constraints. Consider

$$\min_{x \in \mathbb{R}^n} \|x\|_2^2 \quad \text{s.t.} \quad Ax = b,$$

where $A \in \mathbb{R}^{p \times n}$ with $p < n$ and $b \in \mathbb{R}^p$.

(a) Write the Lagrangian $\mathcal{L}(x, \mu)$.

(b) Derive the dual function $q(\mu)$ by solving the stationarity condition $\nabla_x \mathcal{L} = 0$.

(c) Write the dual problem explicitly. Show it is a concave quadratic in μ.

(d) Assuming AA^T is invertible, solve the dual problem and recover the primal optimal solution x^*.

Solution

(a) With multiplier $\mu \in \mathbb{R}^p$ for the equality constraint:

$$\mathcal{L}(x, \mu) = \|x\|^2 + \mu^T(Ax - b) = x^T x + \mu^T Ax - \mu^T b.$$

(b) The Lagrangian is strictly convex in x (Hessian $= 2I \succ 0$). Setting $\nabla_x \mathcal{L} = 2x + A^T\mu = 0$ gives:

$$x = -\frac{1}{2}A^T\mu.$$

Substituting:

$$q(\mu) = \frac{1}{4}\|A^T\mu\|^2 + \mu^T A\left(-\frac{1}{2}A^T\mu\right) - \mu^T b = -\frac{1}{4}\mu^T AA^T\mu - \mu^T b.$$

(c) The dual problem is:

$$\max_{\mu \in \mathbb{R}^p} \; -\frac{1}{4}\mu^T AA^T\mu - \mu^T b.$$

Since $AA^T \succeq 0$, the objective is concave quadratic in μ (unconstrained, since μ is free for equality constraints).

(d) Setting $\nabla_\mu q = -\frac{1}{2}AA^T\mu - b = 0$:

$$\mu^* = -2(AA^T)^{-1}b.$$

Recovering the primal solution: $x^* = -\frac{1}{2}A^T\mu^* = A^T(AA^T)^{-1}b$.
This is exactly the minimum-norm solution to $Ax = b$, also known as the Moore–Penrose pseudoinverse solution $x^* = A^\dagger b$ when A has full row rank.

Exercise 2.3 [Medium]
Dual of a linear program. Consider the LP in inequality form:

$$\min_{x \in \mathbb{R}^n} c^T x \quad \text{s.t.} \quad Ax \le b,$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$.

(a) Write the Lagrangian $\mathcal{L}(x, \lambda)$ with $\lambda \in \mathbb{R}^m$, $\lambda \ge 0$.

(b) Derive the dual function $q(\lambda) = \inf_x \mathcal{L}(x, \lambda)$. Identify the hidden constraint that must hold for $q(\lambda) > -\infty$.

(c) Write the dual problem explicitly, making the hidden constraint explicit. Show that the dual of a linear program is again a linear program.

Solution

(a) The Lagrangian is:

$$\mathcal{L}(x, \lambda) = c^T x + \lambda^T (Ax - b) = (c + A^T \lambda)^T x - \lambda^T b.$$

(b) The Lagrangian is linear (affine) in x. Its infimum over $x \in \mathbb{R}^n$:

$$q(\lambda) = \inf_x \left[(c + A^T \lambda)^T x - \lambda^T b\right] = \begin{cases} -\lambda^T b & \text{if } c + A^T \lambda = 0, \\ -\infty & \text{otherwise.} \end{cases}$$

The hidden constraint is $A^T \lambda + c = 0$, i.e., $A^T \lambda = -c$.
(c) Making the hidden constraint explicit, the dual problem is:

$$\max_{\lambda \in \mathbb{R}^m} \; -b^T \lambda \quad \text{s.t.} \quad A^T \lambda + c = 0, \quad \lambda \geq 0.$$

Equivalently: $\max_{\lambda \geq 0} -b^T \lambda$ s.t. $A^T \lambda = -c$.
This is a linear program (linear objective, linear equality and inequality constraints). Therefore the **dual of an LP is again an LP**. One can also eliminate the equality by writing it as two inequalities, confirming the LP structure.

Exercise 2.4 [Hard]
Dual of the Support Vector Machine. Consider the hard-margin SVM problem with training data $\{(x_i, y_i)\}_{i=1}^N$, where $x_i \in \mathbb{R}^D$ and $y_i \in \{-1, +1\}$:

$$\min_{w \in \mathbb{R}^D} \frac{1}{2} \|w\|_2^2 \quad \text{s.t.} \quad y_i (w^T x_i) \geq 1, \quad i = 1, \ldots, N.$$

(a) Rewrite the constraints in the form $g_i(w) \leq 0$ and write the Lagrangian.

(b) Derive the dual function by minimizing the Lagrangian over w. Express it purely in terms of the multipliers α_i and the data.

(c) Write the dual problem. Show that the dual depends on the data only through the inner products $\langle x_i, x_j \rangle$. Why is this important?

(d) State the complementary slackness conditions and explain why data points with $\alpha_i > 0$ are called *support vectors*.

(e) This problem is convex. Argue why Slater's condition holds if the data is linearly separable, and conclude that strong duality holds.

Solution

(a) Constraints: $g_i(w) = 1 - y_i(w^T x_i) \leq 0$. Lagrangian:

$$\mathcal{L}(w, \alpha) = \frac{1}{2}\|w\|^2 + \sum_{i=1}^{N} \alpha_i (1 - y_i w^T x_i), \quad \alpha_i \geq 0.$$

(b) Minimize over w:

$$\nabla_w \mathcal{L} = w - \sum_{i=1}^{N} \alpha_i y_i x_i = 0 \quad \Longrightarrow \quad w = \sum_{i=1}^{N} \alpha_i y_i x_i.$$

Substitute back:

$$q(\alpha) = \frac{1}{2}\left\|\sum_i \alpha_i y_i x_i\right\|^2 + \sum_i \alpha_i - \sum_i \alpha_i y_i \left(\sum_j \alpha_j y_j x_j\right)^T x_i$$

$$= \sum_{i=1}^{N} \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle.$$

(c) Dual problem:

$$\max_{\alpha \geq 0} \sum_{i=1}^{N} \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle.$$

The dual depends on the data only through the Gram matrix $K_{ij} = \langle x_i, x_j \rangle$. This is the foundation of the **kernel trick**: replacing $\langle x_i, x_j \rangle$ with a kernel function $k(x_i, x_j)$ allows SVMs to operate in implicit high-dimensional feature spaces without computing the features explicitly.

(d) Complementary slackness: $\alpha_i(1 - y_i w^{*T} x_i) = 0$ for all i.

If $\alpha_i > 0$, then $y_i w^{*T} x_i = 1$: the point lies exactly on the margin. These are the *support vectors* — the only training points that influence the decision boundary (since $w^* = \sum_i \alpha_i y_i x_i$ and all other $\alpha_i = 0$). Points with $\alpha_i = 0$ satisfy $y_i w^{*T} x_i > 1$ (strictly inside the correct side) and do not affect the solution.

(e) The objective $\frac{1}{2}\|w\|^2$ is strictly convex. The constraints $1 - y_i w^T x_i \leq 0$ are affine in w, hence convex. If the data is linearly separable, there exists $\hat{w}$ with $y_i \hat{w}^T x_i > 1$ for all i (strict inequality), i.e., the constraints hold strictly. This is exactly Slater's condition (and since all constraints are affine, strict inequality is technically not even needed). Strong duality holds: $p^* = d^*$.

Weak Duality, Strong Duality, and Slater's Condition

Exercise 3.1 [Easy]

Consider the problem

$$\min_{x \in \mathbb{R}} (x - 3)^2 \quad \text{s.t.} \quad x \geq 1.$$

(a) What is the primal optimal value p^*?

(b) Compute the dual function $q(\lambda)$ and evaluate it at $\lambda = 1$. Verify that $q(1) \leq p^*$ (weak duality).

(c) Prove the weak duality theorem in general: for any $\lambda \geq 0$, $q(\lambda) \leq p^*$.

Solution

(a) The unconstrained minimum is at $x = 3$, which satisfies $x \geq 1$. Therefore the constraint is inactive and $p^* = 0$ at $x^* = 3$.
(b) Lagrangian: $\mathcal{L}(x, \lambda) = (x-3)^2 + \lambda(1-x)$. Minimize over x:

$$\frac{\partial \mathcal{L}}{\partial x} = 2(x-3) - \lambda = 0 \implies x = 3 + \frac{\lambda}{2}.$$

Dual function:

$$q(\lambda) = \left(\frac{\lambda}{2}\right)^2 + \lambda\left(1 - 3 - \frac{\lambda}{2}\right) = \frac{\lambda^2}{4} - 2\lambda - \frac{\lambda^2}{2} = -2\lambda - \frac{\lambda^2}{4}.$$

At $\lambda = 1$: $q(1) = -2 - 1/4 = -9/4 \leq 0 = p^*$ ✓.
(c) Let x' be any feasible point (so $g_i(x') \leq 0$, $h_j(x') = 0$). For $\lambda \geq 0$:

$$\mathcal{L}(x', \lambda, \mu) = f(x') + \sum_i \lambda_i \underbrace{g_i(x')}_{\leq 0} + \sum_j \mu_j \underbrace{h_j(x')}_{=0} \leq f(x').$$

Taking infimum over all x (not just feasible): $q(\lambda, \mu) = \inf_x \mathcal{L}(x, \lambda, \mu) \leq \mathcal{L}(x', \lambda, \mu) \leq f(x')$. Since this holds for *every* feasible x', taking the infimum over feasible points: $q(\lambda, \mu) \leq p^*$. □

Exercise 3.2 [Medium]
Consider the problem

$$\min_{x \in \mathbb{R}^2} \; e^{-x_1} \quad \text{s.t.} \quad \frac{x_1^2}{x_2} \leq 0, \quad x_2 > 0.$$

We restrict to the domain $\mathcal{D} = \{x \in \mathbb{R}^2 : x_2 > 0\}$.

(a) Show that the primal optimal value is $p^* = 1$.
(b) Compute the dual function $q(\lambda)$ and the dual optimal value d^*.
(c) Is there a duality gap? Does Slater's condition hold? Explain the connection.

Solution

(a) The constraint $x_1^2/x_2 \leq 0$ with $x_2 > 0$ requires $x_1^2 \leq 0$, which forces $x_1 = 0$. Then the objective is $e^{-0} = 1$ for any $x_2 > 0$. So $p^* = 1$.
(b) Lagrangian on domain $\mathcal{D}$:

$$\mathcal{L}(x, \lambda) = e^{-x_1} + \lambda \frac{x_1^2}{x_2}, \quad \lambda \geq 0,\ x_2 > 0.$$

For the dual function, we take $\inf_{x \in \mathcal{D}} \mathcal{L}$.
For any $\lambda \geq 0$: fix $x_2 > 0$. As $x_1 \to +\infty$, $e^{-x_1} \to 0$ and $\lambda x_1^2/x_2 \to +\infty$. As $x_1 \to -\infty$, $e^{-x_1} \to +\infty$. So we can optimize in x_1.
However, for any fixed $\lambda \geq 0$ and x_1: we can send $x_2 \to +\infty$ to make $\lambda x_1^2/x_2 \to 0$, so the penalty vanishes. Therefore:

$$q(\lambda) = \inf_{x_1, x_2 > 0} \left[e^{-x_1} + \lambda \frac{x_1^2}{x_2} \right] = \inf_{x_1} e^{-x_1} = 0.$$

(Take $x_1 \to +\infty$ and $x_2 \to +\infty$ fast enough.)
Thus $q(\lambda) = 0$ for all $\lambda \geq 0$, and $d^* = 0$.
(c) The duality gap is $p^* - d^* = 1 - 0 = 1 > 0$. Strong duality fails.
Slater's condition requires a strictly feasible point: x with $x_1^2/x_2 < 0$ and $x_2 > 0$. But $x_1^2/x_2 \geq 0$ for $x_2 > 0$, so strict inequality is impossible. Slater's condition does **not** hold, which is consistent with the duality gap. This example shows that Slater's condition, while sufficient, is genuinely needed: without it, even convex problems can have duality gaps.

Exercise 3.3 [Medium]
Consider the convex optimization problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad Ax = b, \quad g_i(x) \leq 0,\ i = 1, \ldots, r,$$

where f and all g_i are convex. Suppose Slater's condition holds (there exists $\tilde{x}$ with $A\tilde{x} = b$ and $g_i(\tilde{x}) < 0$ for all i).

(a) State the consequence of Slater's condition for this problem.

(b) Prove that for a convex problem with strong duality, if the primal optimum x^* and dual optimum (λ^*, μ^*) are attained, then the KKT conditions hold at (x^*, λ^*, μ^*).

(Hint: Use $f(x^*) = p^* = d^* = q(\lambda^*, \mu^*) = \inf_x \mathcal{L}(x, \lambda^*, \mu^*)$ and argue about the chain of inequalities.)

Solution

(a) Slater's condition implies that **strong duality holds**: $d^* = p^*$. Moreover, the dual optimum is attained, i.e., there exist (λ^*, μ^*) with $q(\lambda^*, \mu^*) = d^* = p^*$.
(b) By strong duality and the definitions:

$$f(x^*) = p^* = d^* = q(\lambda^*, \mu^*) = \inf_x \mathcal{L}(x, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*).$$

Expanding the last term:

$$\mathcal{L}(x^*, \lambda^*, \mu^*) = f(x^*) + \sum_i \lambda_i^* g_i(x^*) + \sum_j \mu_j^* h_j(x^*).$$

Since x^* is primal feasible: $g_i(x^*) \leq 0$, $h_j(x^*) = 0$ (here $h_j(x) = a_j^T x - b_j$). With $\lambda^* \geq 0$:

$$\sum_i \lambda_i^* g_i(x^*) \leq 0, \qquad \sum_j \mu_j^* h_j(x^*) = 0.$$

Thus $\mathcal{L}(x^*, \lambda^*, \mu^*) \leq f(x^*)$.
Combining with the previous inequality: $f(x^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*) \leq f(x^*)$, so equality holds throughout. This gives us:
Complementary slackness: $\sum_i \lambda_i^* g_i(x^*) = 0$. Since each term $\lambda_i^* g_i(x^*) \leq 0$ and they sum to zero, each term is zero: $\lambda_i^* g_i(x^*) = 0$ for all i.
Stationarity: Since $\mathcal{L}(x^*, \lambda^*, \mu^*) = \inf_x \mathcal{L}(x, \lambda^*, \mu^*)$, x^* minimizes the Lagrangian over x. For convex $\mathcal{L}(\cdot, \lambda^*, \mu^*)$, this requires $\nabla_x \mathcal{L}(x^*, \lambda^*, \mu^*) = 0$.
Primal and dual feasibility: Hold by assumption (x^* is primal feasible, $\lambda^* \geq 0$).
All four KKT conditions are satisfied. □

Exercise 3.4 [Hard]
When strong duality fails. Consider the non-convex problem

$$\min_{x \in \mathbb{R}} x \quad \text{s.t.} \quad x^2 \leq 0.$$

(a) Find the primal optimal value p^*.
(b) Compute the dual function $q(\lambda)$ and the dual optimal d^*.
(c) Show there is a duality gap and explain why. Which assumption of Slater's theorem is violated?
(d) Now consider the modified problem $\min_x x$ s.t. $x^2 - 1 \leq 0$. Show that Slater's condition holds, compute the dual function and dual optimal value, and verify strong duality.

Solution

(a) The constraint $x^2 \leq 0$ forces $x = 0$, so $p^* = 0$.
(b) Lagrangian: $\mathcal{L}(x, \lambda) = x + \lambda x^2$, $\lambda \geq 0$.
For $\lambda = 0$: $\inf_x x = -\infty$, so $q(0) = -\infty$.
For $\lambda > 0$: minimize the convex quadratic $\lambda x^2 + x$. Setting $2\lambda x + 1 = 0$: $x = -1/(2\lambda)$. Then:

$$q(\lambda) = -\frac{1}{2\lambda} + \lambda \cdot \frac{1}{4\lambda^2} = -\frac{1}{4\lambda}.$$

So $q(\lambda) = -1/(4\lambda)$ for $\lambda > 0$. Taking $\lambda \to \infty$: $q(\lambda) \to 0^-$, so $d^* = \sup_{\lambda>0} q(\lambda) = 0$.
But the supremum is not attained (no finite λ achieves $q(\lambda) = 0$). Actually, note that $d^* = 0$ in the limit but $q(\lambda) < 0$ for all finite $\lambda > 0$. Since $\sup_{\lambda>0}(-1/(4\lambda)) = 0$, we have $d^* = 0 = p^*$.
So in this case there is actually no duality gap. Let us instead consider the problem:
The corrected analysis: $d^* = \sup_{\lambda\geq 0} q(\lambda) = 0$ (the sup approaches 0 but is never attained). Since $p^* = 0$, we have $d^* = p^*$. The duality gap is zero but the dual optimum is not attained.
(c) Although $d^* = p^*$ here, the dual optimum is not attained: there is no $\lambda^* \geq 0$ with $q(\lambda^*) = 0$. The problem is non-convex ($g(x) = x^2$ is convex, but the feasible set $\{x : x^2 \leq 0\} = \{0\}$ makes the problem degenerate). Slater's condition requires strict feasibility: a point $\tilde{x}$ with $\tilde{x}^2 < 0$, which is impossible. Without Slater's condition, we cannot guarantee that the dual optimum is attained, even when $d^* = p^*$.
(d) Modified problem: $\min_x x$ s.t. $x^2 - 1 \leq 0$ (i.e., $-1 \leq x \leq 1$). Primal optimal: $x^* = -1$, $p^* = -1$.
Slater's: take $\tilde{x} = 0$: $0^2 - 1 = -1 < 0$ ✓.
Lagrangian: $\mathcal{L}(x, \lambda) = x + \lambda(x^2 - 1)$. For $\lambda > 0$: $x = -1/(2\lambda)$.

$$q(\lambda) = -\frac{1}{2\lambda} + \lambda\left(\frac{1}{4\lambda^2} - 1\right) = -\frac{1}{4\lambda} - \lambda.$$

Maximize: $q'(\lambda) = \frac{1}{4\lambda^2} - 1 = 0 \Rightarrow \lambda^* = 1/2$.

$$q(1/2) = -\frac{1}{2} - \frac{1}{2} = -1 = p^*.$$

Strong duality holds as guaranteed by Slater's condition. □

Mixed Challenging Exercises

Exercise 4.1 [Hard]
Entropy maximization and KL divergence (Exam-style). Let $p = (p_1, \ldots, p_n)$ be a fixed probability distribution with $p_i > 0$ for all i. Consider the problem

$$\min_{x\in\mathbb{R}^n} \sum_{i=1}^n x_i\big(\log(x_i) - \log(p_i)\big) \quad \text{s.t.} \quad \sum_{i=1}^n x_i = 1, \quad x_i \geq 0\ \forall i.$$

Note: this objective is the KL divergence $D_{\mathrm{KL}}(x\|p)$.

(a) Show that $\phi(x) = \sum_{i=1}^n x_i(\log x_i - \log p_i)$ is strictly convex on $\{x : x_i > 0\}$.

(b) Write the Lagrangian with multiplier μ for the equality constraint and $\lambda_i \geq 0$ for $-x_i \leq 0$. Derive all KKT conditions.

(c) Using complementary slackness and stationarity, show that $x_i^* = p_i$ for all i and determine the multipliers.

(d) What is the optimal value? Explain why this is consistent with the known property $D_{\mathrm{KL}}(q\|p) \geq 0$ with equality iff $q = p$.

(e) Assuming strong duality holds, derive the dual function and solve the dual problem. Verify that the dual optimal value equals the primal optimal value.

(f) *(Bonus)* The problem also has a saddle-point interpretation: $p^* = \inf_x \sup_{\lambda \geq 0, \mu} \mathcal{L}(x, \lambda, \mu)$. Without proof, state the corresponding expression for d^* and the condition under which they are equal.

Solution

(a) For $x_i > 0$, the function $\varphi_i(x_i) = x_i(\log x_i - \log p_i)$ has second derivative:

$$\varphi_i''(x_i) = \frac{1}{x_i} > 0.$$

Since $\phi(x) = \sum_i \varphi_i(x_i)$ is a sum of strictly convex functions of independent variables, it is strictly convex. Equivalently, the Hessian $H\phi(x) = \mathrm{diag}(1/x_1, \ldots, 1/x_n) \succ 0$ for $x_i > 0$.
(b) Lagrangian:

$$\mathcal{L}(x, \lambda, \mu) = \sum_i x_i(\log x_i - \log p_i) + \mu \left(\sum_i x_i - 1 \right) + \sum_i \lambda_i(-x_i).$$

KKT conditions:

- Stationarity: $\log x_i - \log p_i + 1 + \mu - \lambda_i = 0$ for all i
- Primal feasibility: $\sum_i x_i = 1$, $x_i \geq 0$
- Dual feasibility: $\lambda_i \geq 0$
- Complementary slackness: $\lambda_i x_i = 0$ for all i

(c) Suppose $x_i > 0$. By complementary slackness, $\lambda_i = 0$. Stationarity gives:

$$\log x_i = \log p_i - 1 - \mu \quad \Rightarrow \quad x_i = p_i \cdot e^{-(1+\mu)}.$$

Using $\sum_i x_i = 1$: $e^{-(1+\mu)} \sum_i p_i = 1$. Since $\sum_i p_i = 1$, we get $e^{-(1+\mu)} = 1$, i.e., $\mu = -1$. Thus $x_i = p_i$ for all i.
If any $x_j = 0$: stationarity gives $\lambda_j = \log(0) - \log p_j + 1 + \mu \to -\infty$, violating $\lambda_j \geq 0$. So all $x_i > 0$ and the unique solution is $x^* = p$.
Multipliers: $\mu^* = -1$, $\lambda_i^* = 0$ for all i.
(d) Optimal value: $\phi(p) = \sum_i p_i(\log p_i - \log p_i) = 0$. This is consistent with Gibbs' inequality: $D_{\mathrm{KL}}(q\|p) \geq 0$ for all distributions q, with equality if and only if $q = p$. Our optimization confirms that p is the unique minimizer.
(e) From stationarity: $x_i = p_i e^{\lambda_i - 1 - \mu}$. Substituting into the Lagrangian (after simplification):

$$q(\lambda, \mu) = -e^{-1-\mu} \sum_i p_i e^{\lambda_i} + \mu + \sum_i \lambda_i \cdot 0 - \mu.$$

More carefully: substituting $x_i = p_i e^{\lambda_i - 1 - \mu}$ into $\mathcal{L}$:
After computation, the dual function simplifies to:

$$q(\mu) = -e^{-(1+\mu)} - \mu \quad (\text{with } \lambda_i = 0).$$

Maximize: $q'(\mu) = e^{-(1+\mu)} - 1 = 0 \Rightarrow \mu^* = -1$, giving $q(-1) = -1 + 1 = 0 = p^*$.
Strong duality holds: $d^* = p^* = 0$.
(f) The saddle-point interpretation gives $d^* = \sup_{\lambda \geq 0, \mu} \inf_x \mathcal{L}(x, \lambda, \mu)$. Under strong duality:

$$\sup_{\lambda \geq 0, \mu} \inf_x \mathcal{L}(x, \lambda, \mu) = \inf_x \sup_{\lambda \geq 0, \mu} \mathcal{L}(x, \lambda, \mu) = p^*.$$

The min and max can be exchanged when strong duality holds.

Exercise 4.2 [Hard]

Constrained quadratic optimization with multiple constraints. Consider

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^T Q x + c^T x \quad \text{s.t.} \quad Ax \leq b, \quad Cx = d,$$

where $Q \in \mathbb{R}^{n \times n}$ is symmetric positive definite, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{p \times n}$, $d \in \mathbb{R}^p$.

(a) Write the Lagrangian $\mathcal{L}(x, \lambda, \mu)$ with $\lambda \in \mathbb{R}^m_+$ and $\mu \in \mathbb{R}^p$.

(b) Derive the dual function $q(\lambda, \mu)$ in closed form.

(c) Write the dual problem. Explain why it is always a concave maximization problem, regardless of the original problem's convexity.

(d) Prove that Slater's condition holds if there exists $\tilde{x}$ with $A\tilde{x} < b$ (componentwise) and $C\tilde{x} = d$. Conclude strong duality.

(e) For the specific instance $n = 2$, $Q = I$, $c = 0$, with constraints $x_1 + x_2 \leq 1$ and $x_1 - x_2 = 0$: use the KKT conditions to find the optimal solution, the active constraints, and all multipliers.

Solution

(a) The Lagrangian is:

$$\mathcal{L}(x, \lambda, \mu) = \frac{1}{2}x^T Qx + c^T x + \lambda^T (Ax - b) + \mu^T (Cx - d).$$

(b) Since $Q \succ 0$, $\mathcal{L}$ is strictly convex in x. Setting $\nabla_x \mathcal{L} = 0$:

$$Qx + c + A^T\lambda + C^T\mu = 0 \quad \Longrightarrow \quad x = -Q^{-1}(c + A^T\lambda + C^T\mu).$$

Substituting back (let $v = c + A^T\lambda + C^T\mu$):

$$q(\lambda, \mu) = -\frac{1}{2}v^T Q^{-1} v - \lambda^T b - \mu^T d.$$

Expanded:

$$q(\lambda, \mu) = -\frac{1}{2}(c + A^T\lambda + C^T\mu)^T Q^{-1}(c + A^T\lambda + C^T\mu) - b^T\lambda - d^T\mu.$$

(c) Dual problem: $\max_{\lambda \geq 0, \mu} \; q(\lambda, \mu)$.
The dual function $q(\lambda, \mu)$ is always concave because it is defined as $q(\lambda, \mu) = \inf_x \mathcal{L}(x, \lambda, \mu)$, i.e., a pointwise infimum of affine functions of (λ, μ) (for each fixed x, $\mathcal{L}$ is affine in (λ, μ)). The pointwise infimum of affine (hence concave) functions is concave. This holds regardless of whether the primal is convex.
(d) Slater's condition for convex problems requires: (i) the problem is convex, and (ii) there exists a strictly feasible point. Here $f(x) = \frac{1}{2}x^T Qx + c^T x$ is convex ($Q \succ 0$), the constraints $Ax \leq b$ are affine, and $Cx = d$ is affine. If $\tilde{x}$ exists with $A\tilde{x} < b$ (strict) and $C\tilde{x} = d$, Slater's condition holds. (For affine inequality constraints, one only needs feasibility, not strict feasibility, but here we are given strict feasibility.) Therefore $d^* = p^*$.
(e) Problem: $\min \frac{1}{2}(x_1^2 + x_2^2)$ s.t. $g(x) = x_1 + x_2 - 1 \leq 0$, $h(x) = x_1 - x_2 = 0$.
KKT stationarity: $x_1 + \lambda + \mu = 0$ and $x_2 + \lambda - \mu = 0$.
From $x_1 = x_2$ (equality constraint): let $x_1 = x_2 = t$. Then: $t + \lambda + \mu = 0$ and $t + \lambda - \mu = 0$.
Subtracting: $2\mu = 0 \Rightarrow \mu = 0$. So $t = -\lambda$.
Case 1: $\lambda = 0$ (constraint inactive). Then $t = 0$, $x^* = (0, 0)$. Check: $0 + 0 = 0 \leq 1$ ✓. So $x^* = (0, 0)$ with $\lambda^* = 0$, $\mu^* = 0$, and optimal value $= 0$.
The constraint $x_1 + x_2 \leq 1$ is inactive (slack $= 1$), which is consistent with $\lambda^* = 0$ by complementary slackness. The equality constraint $x_1 = x_2$ is active (as it must be), with $\mu^* = 0$.

Exercise 4.3 [Hard]
Maximum likelihood with constraints. Consider N observations $y_1, \ldots, y_N$ drawn i.i.d. from a categorical distribution over $\{1, \ldots, K\}$ with probabilities $\theta = (\theta_1, \ldots, \theta_K)$. Let $n_k = |\{i : y_i = k\}|$ be the count of category k. The log-likelihood is

$$\ell(\theta) = \sum_{k=1}^{K} n_k \log \theta_k.$$

(a) Write the constrained MLE problem as a maximization with constraints $\sum_k \theta_k = 1$ and $\theta_k \geq 0$.

(b) Convert to a minimization problem. Write the Lagrangian and KKT conditions.

(c) Using complementary slackness, show that the MLE is $\hat{\theta}_k = n_k/N$.

(d) Derive the dual function and dual problem. Verify strong duality by showing the optimal dual value equals $-\ell(\hat{\theta})$.

(e) The dual of MLE connects to exponential families. Show that the stationarity condition can be written as $\mathbb{E}_{\hat{\theta}}[\mathbf{1}_{Y=k}] = n_k/N$, i.e., the model's expected sufficient statistics match the empirical ones. This is the *moment matching* property of MLEs in exponential families.

Solution

(a) The MLE problem:

$$\max_{\theta \in \mathbb{R}^K} \sum_{k=1}^{K} n_k \log \theta_k \quad \text{s.t.} \quad \sum_{k=1}^{K} \theta_k = 1, \quad \theta_k \geq 0 \; \forall k.$$

(b) Equivalent minimization: $\min_\theta - \sum_k n_k \log \theta_k$ s.t. $\sum_k \theta_k = 1$, $-\theta_k \leq 0$.
Lagrangian:

$$\mathcal{L}(\theta, \lambda, \mu) = -\sum_k n_k \log \theta_k + \mu \left(\sum_k \theta_k - 1 \right) + \sum_k \lambda_k (-\theta_k).$$

KKT conditions:

- Stationarity: $-n_k/\theta_k + \mu - \lambda_k = 0$ for all k
- Primal feasibility: $\sum_k \theta_k = 1$, $\theta_k \geq 0$
- Dual feasibility: $\lambda_k \geq 0$
- Complementary slackness: $\lambda_k \theta_k = 0$ for all k

(c) Assume $n_k > 0$ for all k (each category observed at least once). If $\theta_k > 0$, then $\lambda_k = 0$ by complementary slackness. Stationarity: $\theta_k = n_k/\mu$. If $\theta_k = 0$, then $-n_k/0$ diverges, which is impossible for $n_k > 0$. So all $\theta_k > 0$.
Using $\sum_k \theta_k = \sum_k n_k/\mu = N/\mu = 1$, so $\mu = N$. Therefore:

$$\hat{\theta}_k = \frac{n_k}{N}.$$

(d) From stationarity with $\lambda_k = 0$: $\theta_k = n_k/\mu$. Substituting into Lagrangian:

$$\begin{aligned} q(\mu) &= -\sum_k n_k \log(n_k/\mu) + \mu \left(\sum_k n_k/\mu - 1 \right) \\ &= -\sum_k n_k \log n_k + N \log \mu + N - \mu. \end{aligned}$$

Maximize over μ: $q'(\mu) = N/\mu - 1 = 0 \Rightarrow \mu^* = N$.

$$d^* = q(N) = -\sum_k n_k \log n_k + N \log N = -\sum_k n_k \log(n_k/N) = -\ell(\hat{\theta}).$$

Primal optimal: $p^* = -\ell(\hat{\theta}) = -\sum_k n_k \log(n_k/N)$. So $d^* = p^*$, confirming strong duality.
(e) The stationarity condition $\theta_k = n_k/\mu$ with $\mu = N$ gives $\hat{\theta}_k = n_k/N$. Under the model $\hat{\theta}$:

$$\mathbb{E}_{\hat{\theta}}[\mathbf{1}_{Y=k}] = P_{\hat{\theta}}(Y = k) = \hat{\theta}_k = \frac{n_k}{N} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}_{y_i = k}.$$

The left side is the model expectation; the right side is the empirical frequency. The MLE equates these, which is the **moment matching** property: the MLE in an exponential family is found by setting the expected sufficient statistics equal to the observed sufficient statistics.